

SAITEJA RAVIPATI

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Professional Summary

Results-driven Data Engineer with ~5 years of experience in designing and delivering scalable ETL pipelines, real-time streaming workflows, and cloud-native data platforms across AWS, Snowflake, and Spark Structured Streaming. Skilled in architecting end-to-end batch and streaming pipelines, developing reusable Python/PySpark frameworks, and embedding CI/CD automation using Azure DevOps, Bamboo, and GitHub Actions. Experienced in Snowflake performance tuning, cost optimization, and data quality frameworks, ensuring high accuracy and efficiency in enterprise analytics. Proficient in leveraging MCP (Model Context Protocol) and AI copilots to automate bug triage, validation, and pipeline development, reducing manual effort by 30%+ and accelerating release cycles by 20%+. Recognized for delivering secure, reliable, and business-aligned data solutions that drive ROI, operational efficiency, and data-driven decision-making at enterprise scale.

Skills

Languages & Scripting

Python, SQL, Java, ABAP, SAP SQLScript, Gherkin

Cloud & Platforms

AWS (S3, Glue, Lambda, RDS, EC2, CloudWatch, Kinesis, SQS, SNS) · Azure (DevOps, Functions, Blob Storage) · SAP HANA

ETL, Orchestration & Frameworks

Apache Airflow · AWS Glue · Apache Spark · PySpark · Spark Structured Streaming · Snowpipe · Pandas · Pipeline Framework · SAP HANA SDI

Data Warehousing & Analytics

Snowflake (SQL optimization, warehouse tuning, data sharing, zero-copy cloning) · SAP HANA · Redshift (legacy) · Azure Synapse (exposure)

Big Data & Streaming

Kafka · Spark Streaming · AWS Kinesis · SQS/SNS · SAP HANA Streaming Analytics

Databases

Snowflake · PostgreSQL · Oracle · SAP HANA · AWS RDS

DevOps & CI/CD

Azure DevOps · GitHub Actions · Jenkins ·

Bamboo · Docker · JFrog Artifactory

Visualization & Reporting

Power BI · Tableau · SAP Analytics Cloud

Other Tools & Practices

MCP (Model Context Protocol) · GitHub Copilot

Agent · REST APIs · JIRA · JSON · XML · Linux ·

Test Automation (PyTest, Requests) · Schema Drift

Detection · Anomaly Detection

Work History

Python/AWS Data Engineer, 05/2024 to Current

CaterpillarInc – Irving, TX

At Caterpillar, I work as part of an enterprise data engineering team building large-scale dealer and customer data pipelines. My role emphasizes **pipeline design, reusable frameworks, and AI-assisted automation** to support analytics and operational systems.

- Designed **ETL mappings and transformations** for structured/semi-structured dealer and customer data, leveraging Python (Pandas) and PySpark.
- Developed **reusable Python frameworks** to handle ingestion, transformations, and reporting logic across multiple pipelines.
- Applied **MCP (Model Context Protocol)** to integrate AI copilots into engineering workflows, enabling context-aware bug triaging, log analysis, and automated diagnostics.
- Automated **bug triaging and anomaly detection** using AI + MCP-driven pipelines with CloudWatch integration, reducing mean-time-to-recovery by 20%.
- Leveraged **GitHub Copilot Agent alongside MCP** to auto-generate pipeline scripts, regression logic, and unit tests — reducing manual coding effort and accelerating delivery cycles by 25%.
- Worked closely with **cross-functional teams (QA, DevOps, Analytics)** to gather requirements and coordinate bug triage, ensuring smooth pipeline delivery and faster issue resolution.
- Implemented **CI/CD workflows (Azure DevOps, Bamboo, GitHub Actions)** to embed pipeline validation checkpoints before production releases.
- Optimized **Snowflake SQL transformations and warehouse tuning** (caching, clustering strategies), improving query performance by ~20% while reducing compute costs.
- Scaled **pipeline orchestration for near real-time analytics**, integrating batch + streaming workflows to support operational dashboards with higher reliability.
- Designed **end-to-end pipeline workflows** including mapping, business logic, and error-handling frameworks to improve reliability and reduce rework.
- Implemented **pipeline monitoring and logging layers** to proactively track data quality, health, and SLA adherence, enabling faster troubleshooting and improved operational visibility.

Key Achievements

- Saved **120+ engineering hours/quarter** via Copilot + MCP–assisted pipeline automation.
- Achieved **~30% ROI** by standardizing frameworks and AI-driven workflows across multiple pipelines.
- Improved **pipeline uptime by 20%** with MCP-powered anomaly detection and automated recovery.
- Reduced **release timelines by 25%** by embedding validation into CI/CD workflows and automating pipeline delivery.

Data Engineer, 02/2023 to 03/2024

Autodesk – San Rafael, CA

At Autodesk, I contributed to building **scalable, cloud-native data pipelines** to support enterprise analytics and machine learning initiatives. My focus was on **pipeline engineering, validation frameworks, and CI/CD automation** to deliver reliable, production-ready data solutions.

- Designed and developed **batch and streaming data pipelines** to integrate diverse data sources into analytics-ready models, ensuring scalability and consistency.
- Created **pipeline mappings, business logic, and transformation rules** to streamline ingestion-to-reporting workflows and reduce reprocessing overhead.
- Built **regression and validation frameworks** using Python and PySpark, introducing schema drift detection to reduce manual QA effort by **~35%**.
- Developed **error-handling and reconciliation logic** within pipelines, ensuring data quality and auditability across ingestion, transformation, and load stages.
- Implemented **CI/CD automation (Azure DevOps, Jenkins, GitHub)** with embedded tests and approvals, reducing deployment timelines by **~20%**.
- Optimized **Snowflake SQL queries and transformations** to improve dashboard performance and reduce compute usage.
- Collaborated with **cross-functional teams (Product, Analytics, QA, DevOps)** to align pipeline workflows with business SLAs and reporting requirements.
- Enhanced **monitoring and operational visibility** by embedding logging and validation hooks into pipelines, improving defect detection by **~30%**.

Key Achievements

- Reduced **manual QA effort by 35%** through automated regression and schema drift validation.
- Improved **pipeline stability and defect detection by 30%**, minimizing downstream production issues.
- Accelerated **deployment timelines by 20%** with CI/CD-enabled pipeline workflows.
- Delivered **high-quality data pipelines** that directly supported enterprise analytics and ML-driven decision-making.

Software Engineer, 08/2019 to 06/2021

APSPDCL – Andhra Pradesh, India

At APSPDCL, I was part of the data engineering team responsible for **modernizing utility data pipelines** to support operational, billing, and reporting systems. My work focused on **pipeline design, validation logic, and data integration** to ensure accuracy and reliability of critical energy data.

- Designed and implemented **ETL mappings and transformation logic** to process energy consumption and billing data, ensuring accuracy and consistency across enterprise systems.
- Built and maintained **data pipelines** to process metering and grid data, improving operational visibility and outage management.
- Developed **validation and reconciliation logic** to identify discrepancies in billing and consumption data, reducing errors in financial reporting.
- Designed **centralized reporting models and dashboards** to provide insights into grid performance, outages, and consumption trends.
- Collaborated with **cross-functional teams (Operations, Analytics, Reporting)** to define pipeline requirements and streamline data flows across legacy and modern systems.
- Implemented **data quality checks and monitoring** within pipelines to improve trust in analytics and reporting outputs.
- Supported the organization's **digitization initiative**, laying the foundation for future adoption of cloud-based data platforms and scalable analytics.

Key Achievements

- Ensured **high accuracy in billing and energy consumption data**, reducing disputes and improving operational trust.
- Improved **real-time visibility of grid performance**, enabling faster outage response.
- Reduced **reporting delays** by integrating multiple operational data sources into streamlined pipelines.
- Contributed to APSPDCL's **digital modernization roadmap**, setting the stage for advanced analytics and cloud migration.

Education

Master of Science: Information Technology, 12/2022

University of Memphis - Memphis, TN